$$\frac{x^2-3x-18}{x^3-12x^2+32x} \leq 0 \qquad \frac{N}{D} \leq 0$$

$N \geq 0 \quad x^2-3x-18 \geq 0$

$D > 0 \quad x^3-12x^2+32x > 0$

$N \geq 0 \quad x^2-3x-18=0$

$$x_{1,2} = \frac{3 \pm \sqrt{9+72}}{2} \begin{cases} 6 \\ -3 \end{cases}$$

$x \leq -3 \vee x \geq 6$

$x^3-12x^2+32x = x(x^2-12x+32)$

$D > 0 \quad x(x^2-12x+32) > 0$

$x > 0 \quad x^2-12x+32 > 0$

$x_{1,2} = 6 \pm \sqrt{36-32} =$

$x_1 = 8 \quad x_2 = 4$

$x < 4 \vee x > 8$

$-3 \quad 0 \quad 4 \quad 6 \quad 8$



$x \leq -3 \quad 0 < x < 4 \quad 6 \leq x < 8$

$|x| < a$ se $a \leq 0$ impossível.

se $a > 0$ $\quad -a < x < a$

$$\begin{cases} x > 0 & x < a \\ x < 0 & -x < a \Rightarrow x > -a \end{cases}$$

$|2x - 3| < 4$

$-4 < 2x - 3 < 4$

$-1 < 2x < 7$

$-\frac{1}{2} < x < \frac{7}{2}$

$|x| > a$ se $a \leq 0$ $\forall x \in \mathbb{R}$ $(x \neq 0)$

se $a > 0$

$$\begin{cases} x > 0 & x > a \\ x < 0 & -x > a \Rightarrow x < -a \end{cases}$$

$x < -a \ \vee \ x > a$

$|x + 4| > 2$

$x + 4 < -2 \ \vee \ x + 4 > 2$

$x < -6 \ \vee \ x > -2$

$\sqrt[n]{A(x)} > B(x)$ $\sqrt[n]{A(x)} < B(x)$

n dispari $\sqrt[3]{x+3} > x$

$x+3 > x^3$

n pari $(n=2)$

(A) $\sqrt{A(x)} > B(x)$

(1) $\begin{cases} B(x) \geq 0 \\ A(x) > (B(x))^2 \end{cases}$ (2) $\begin{cases} B(x) < 0 \\ A(x) \geq 0 \end{cases}$

$\sqrt{A(x)} < B(x)$

(B) $B(x) < 0$ impossibile.

$\begin{cases} B(x) > 0 \\ A(x) \geq 0 \\ A(x) < (B(x))^2 \end{cases}$

$$2x+1 > \sqrt{4x^2-9x+2}$$

$$\sqrt{4x^2-9x+2} < \overbrace{2x+1}^{B(x)}$$

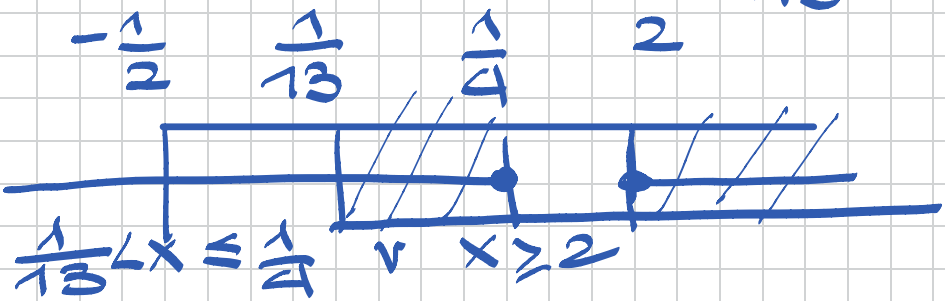
$$\begin{cases} 2x+1 > 0 \\ 4x^2-9x+2 \geq 0 \\ 4x^2-9x+2 < (2x+1)^2 \end{cases}$$

$$4x^2-9x+2=0$$

$$x_{1,2} = \frac{9 \pm \sqrt{81-32}}{8} = \begin{cases} 2 \\ \frac{1}{4} \end{cases}$$

$$\begin{cases} x > -\frac{1}{2} \\ x \leq \frac{1}{4} \vee x \geq 2 \\ \cancel{4x^2}-9x+2 < \cancel{4x^2}+4x+1 \end{cases}$$

$$\begin{cases} x > -\frac{1}{2} \\ x \leq \frac{1}{4} \vee x \geq 2 \\ -13x < -1 \Rightarrow x > \frac{1}{13} \end{cases}$$



$$\frac{1}{13} < x \leq \frac{1}{4} \vee x \geq 2$$

$$0 < \frac{-1+\sqrt{x^2-2x}}{2x} < 1$$

$$\begin{cases} \dfrac{-1+\sqrt{x^2-2x}}{2x} < 1 & \text{(A)} \\ \dfrac{-1+\sqrt{x^2-2x}}{2x} > 0 & \text{(B)} \end{cases}$$

(A) $$\frac{-1+\sqrt{x^2-2x}-2x}{2x} < 0$$

CE $x^2-2x \geq 0$

$x \leq 0 \vee x \geq 2$

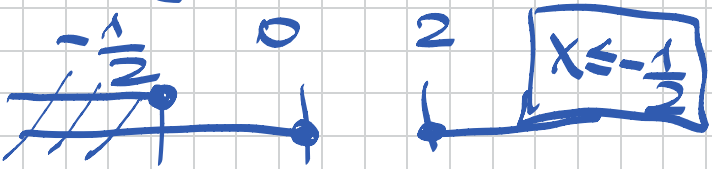
$N>0$ $-1+\sqrt{x^2-2x}-2x>0$

$D>0$ $x>0$

$N>0$ $\sqrt{x^2-2x} > 1+2x$

① $\begin{cases} 1+2x \leq 0 \\ x \leq 0 \quad x \geq 2 \end{cases}$ ② $\begin{cases} 1+2x>0 \\ x^2-2x > (1+2x)^2 \end{cases}$

① $\begin{cases} x \leq -1/2 \\ x \leq 0 \quad x \geq 2 \end{cases}$ ② $\begin{cases} x > -\frac{1}{2} \\ x^2-2x > 1+4x+4x^2 \end{cases}$

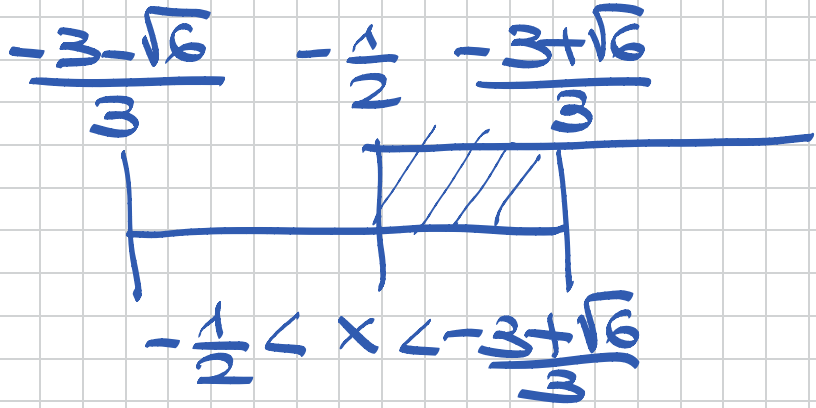


② $\begin{cases} x > -\frac{1}{2} \\ -3x^2 - 6x - 1 > 0 \end{cases}$ $\begin{cases} x > -\frac{1}{2} \\ 3x^2 + 6x + 1 < 0 \end{cases}$

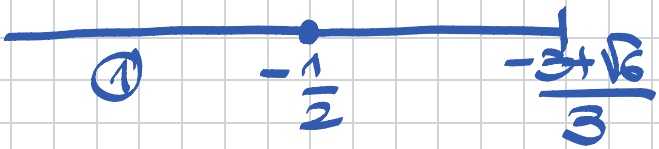
$3x^2 + 6x + 1 = 0$

$x_{1,2} = \frac{-3 \pm \sqrt{9-3}}{3} = \begin{cases} \frac{-3-\sqrt{6}}{3} \\ \frac{-3+\sqrt{6}}{3} \end{cases}$

$\begin{cases} x > -\frac{1}{2} \\ \frac{-3-\sqrt{6}}{3} < x < \frac{-3+\sqrt{6}}{3} \end{cases}$

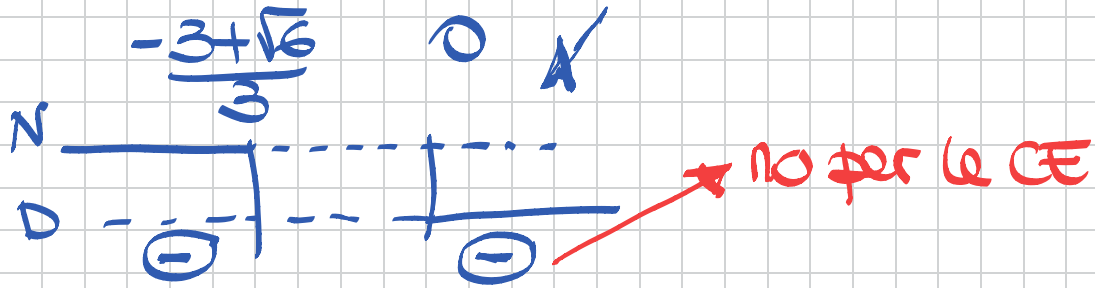


$-\frac{1}{2} < x < \frac{-3+\sqrt{6}}{3}$

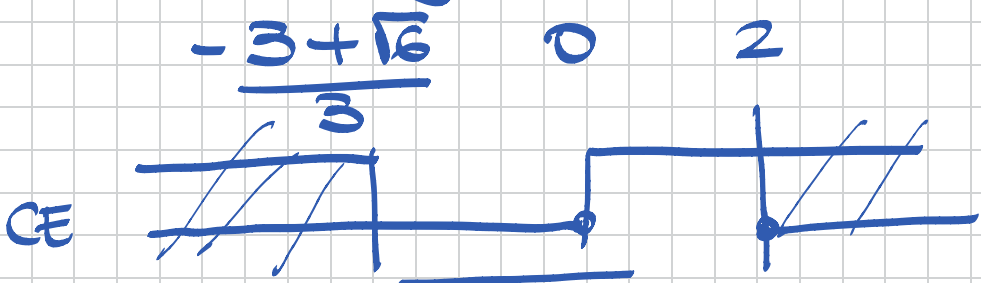


$\Rightarrow N > 0 \qquad x < \frac{-3+\sqrt{6}}{3}$

$D > 0 \qquad x > 0$



$x < \frac{-3+\sqrt{6}}{3} \quad x \geq 2$ (A)



(B) $\frac{-1+\sqrt{x^2-2x}}{2x} > 0$

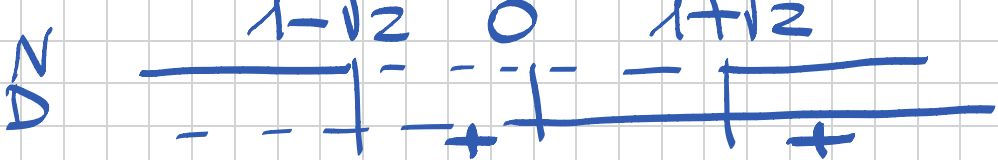
$N > 0 \quad -1+\sqrt{x^2-2x} > 0$

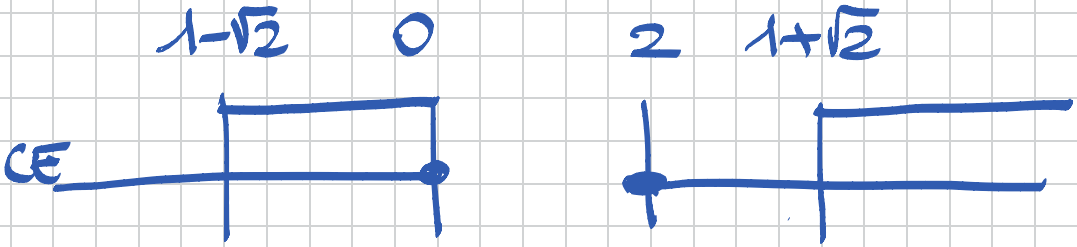
$D > 0 \quad x > 0$

$\sqrt{x^2-2x} > 1$

$x^2-2x > 1 \qquad x^2-2x-1 > 0$

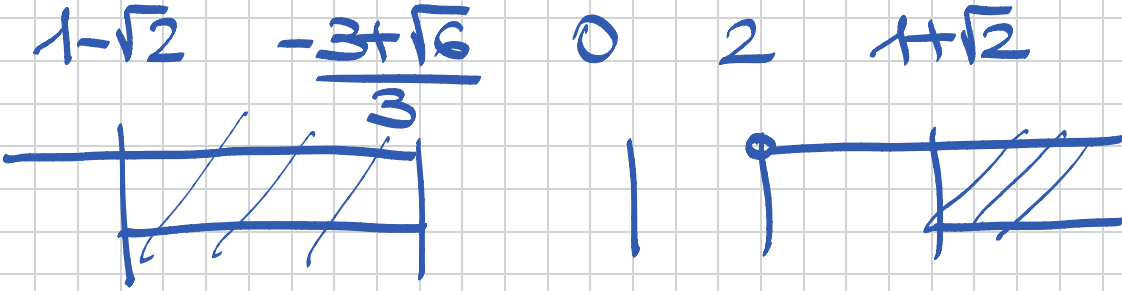
$x_{1,2} = 1 \pm \sqrt{2} \qquad x < 1-\sqrt{2} \vee x > 1+\sqrt{2}$





$\Rightarrow \quad 1-\sqrt{2} < x < 0 \quad \vee \quad x > 1+\sqrt{2}$

$$\begin{cases} x < -\frac{3+\sqrt{6}}{3} \quad x \geq 2 & (A) \\ 1-\sqrt{2} < x < 0 \quad \vee \quad x > 1+\sqrt{2} & (B) \end{cases}$$



$\Rightarrow$ soluzioni: $1-\sqrt{2} < x < -\frac{3+\sqrt{6}}{3}$

$x > 1+\sqrt{2}$

$$\frac{e^{2x}-e^{x}}{2e^{2x}-5e^{x}+2} > -1$$

$$\frac{e^{2x}-e^{x}+2e^{2x}-5e^{x}+2}{2e^{2x}-5e^{x}+2} > 0$$

$$\frac{3e^{2x}-6e^{x}+2}{2e^{2x}-5e^{x}+2}>0$$

$N>0 \quad 3e^{2x}-6e^{x}+2>0$

$3e^{2x}-6e^{x}+2=0$

$t=e^{x} \quad 3t^{2}-6t+2=0$

$$t_{1,2}=\frac{3\pm\sqrt{9-6}}{3} \begin{cases} \frac{3-\sqrt{3}}{3} \\ \frac{3+\sqrt{3}}{3} \end{cases}$$

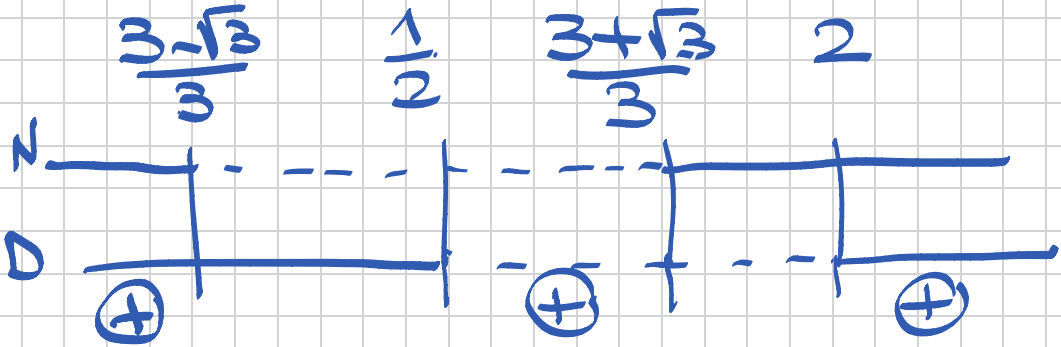
$$t<\frac{3-\sqrt{3}}{3} \vee t>\frac{3+\sqrt{3}}{3}$$

$D>0 \quad 2t^{2}-5t+2>0$

$2t^{2}-5t+2=0$

$$t_{1,2}=\frac{5\pm 3}{4} \begin{cases} 2 \\ \frac{1}{2} \end{cases}$$

$$t<\frac{1}{2} \vee t>2$$



$$t < \frac{3-\sqrt{3}}{3} \vee \frac{1}{2} < t < \frac{3+\sqrt{3}}{3} \vee t > 2$$

$$e^x < \frac{3-\sqrt{3}}{3} \vee \frac{1}{2} < e^x < \frac{3+\sqrt{3}}{3} \vee e^x > 2$$

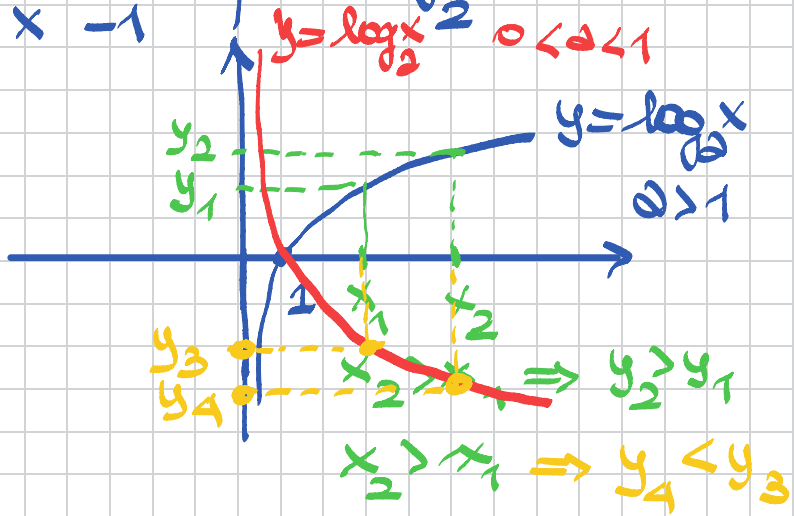
$$x < \log\left(\frac{3-\sqrt{3}}{3}\right)$$

$$\log\frac{1}{2} < x < \log\left(\frac{3+\sqrt{3}}{3}\right)$$

$$x > \log 2$$

$$\log_{\frac{1}{2}}\left(\frac{\sqrt[3]{x}+2}{\sqrt[3]{x}-1}\right) \geq \log_{\frac{1}{2}}\left(\sqrt[3]{x}+3\right)$$

$t = \sqrt[3]{x}$

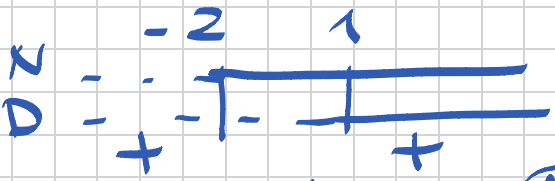


$$\log_{\frac{1}{2}}\left(\frac{t+2}{t-1}\right) \geq \log_{\frac{1}{2}}(t+3)$$

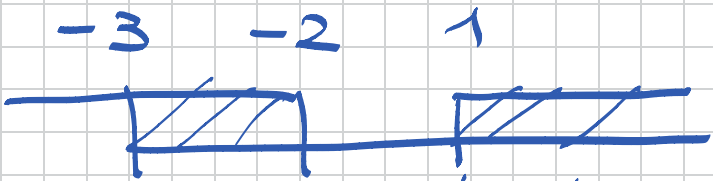
$$\begin{cases} \frac{t+2}{t-1} > 0 & \text{(A)} \\ t+3 > 0 & \text{(B)} \end{cases}$$

(A) $\frac{t+2}{t-1} > 0$ $\quad N>0 \ t>-2$ $\quad D>0 \ t>1$

$t<-2 \vee t>1$



$$\begin{cases} t<-2 \vee t>1 & \text{(A)} \\ t>-3 & \text{(B)} \end{cases}$$



CE : $-3<t<-2 \vee t>1$

$$\frac{t+2}{t-1} \leq t+3$$

$$\frac{t+2-(t+3)(t-1)}{t-1} \leq 0$$

$$\frac{t+2-t^2+t-3t+3}{t-1} \leq 0$$

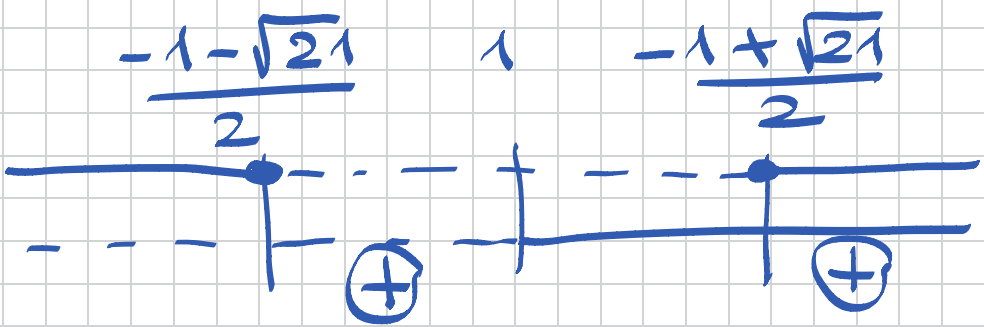
$$\frac{-t^2-t+5}{t-1} \le 0 \qquad \frac{t^2+t-5}{t-1} \ge 0$$

$N \ge 0 \quad t^2+t-5 \ge 0$

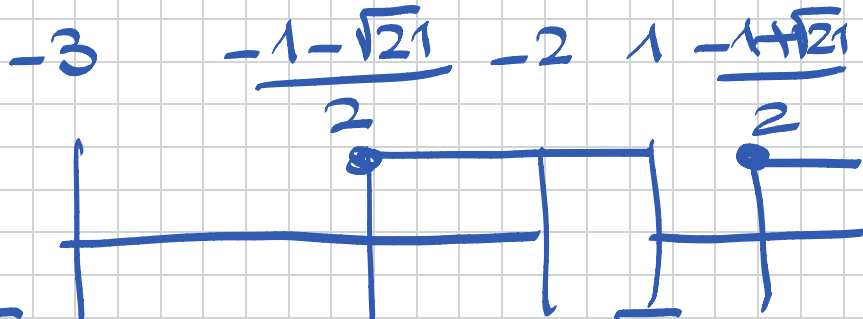
$$t_{1,2} = \frac{-1 \pm \sqrt{21}}{2}$$

$$t \le \frac{-1-\sqrt{21}}{2} \quad \vee \quad t \ge \frac{-1+\sqrt{21}}{2}$$

$D > 0 \quad t > 1$



$$\frac{-1-\sqrt{21}}{2} \le t < 1 \quad \vee \quad t \ge \frac{-1+\sqrt{21}}{2}$$



$$\Rightarrow \frac{-1-\sqrt{21}}{2} < t < -2 \quad \vee \quad t > \frac{-1+\sqrt{21}}{2}$$

$$-\frac{1-\sqrt{21}}{2} \leq \sqrt[3]{x} < -2 \qquad \sqrt[3]{x} > -\frac{1+\sqrt{21}}{2}$$

$$\left(-\frac{1-\sqrt{21}}{2}\right)^3 \leq x < -8 \qquad x > \left(-\frac{1+\sqrt{21}}{2}\right)^3$$

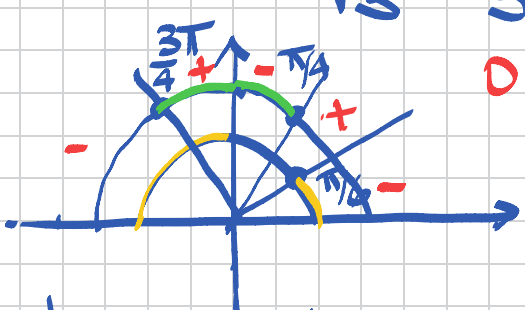
$$\frac{\tan^2 x - \sqrt{3}\tan x}{\tan^2 x - 1} < 1 \qquad 0 \leq x < \pi$$

$$\frac{\tan^2 x - \sqrt{3}\tan x - \tan^2 x + 1}{\tan^2 x - 1} < 0$$

$N > 0 \quad -\sqrt{3}\tan x + 1 > 0$

$D > 0 \quad \tan^2 x - 1 > 0$

$N > 0 \quad \tan x < \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \quad x = \frac{\pi}{6}$



$0 \leq x < \frac{\pi}{6}$

$\frac{\pi}{4} < x < \frac{\pi}{2}$

$\frac{3\pi}{4} < x < \pi$

$D > 0 \quad \tan x = \pm 1$

$\tan x < -1 \vee \tan x > 1$

$\tan x = -1 \quad x = \frac{3}{4}\pi \qquad \tan x = 1 \quad x = \pi/4$

$$\frac{4\sin^2 x-3}{\sin x} > \frac{2\cos x}{|\sin x|}$$

① $\sin x > 0$ $\quad \boxed{0 < x < \pi}$

$$\frac{4\sin^2 x-3}{\sin x} > \frac{2\cos x}{\sin x}$$

$$\frac{4\sin^2 x-3-2\cos x}{\sin x} > 0$$

$N > 0 \quad 4\sin^2 x - 3 - 2\cos x > 0$

$D > 0 \quad \sin x > 0$

$$4(1-\cos^2 x) - 3 - 2\cos x > 0$$

$$-4\cos^2 x - 2\cos x + 1 > 0$$

$$4\cos^2 x + 2\cos x - 1 < 0$$

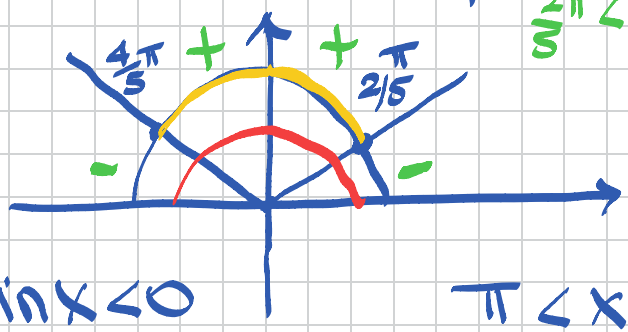
$$\cos x = \frac{-1 \pm \sqrt{5}}{4}$$

$\cos x = -\frac{1-\sqrt{5}}{4}$ $\quad x = \frac{4}{5}\pi ; \frac{6}{5}\pi$

$\cos x = -\frac{1+\sqrt{5}}{4}$ $\quad x = \frac{2}{5}\pi ; \frac{8}{5}\pi$

$$\frac{-1-\sqrt{5}}{4} < \cos x < \frac{-1+\sqrt{5}}{4}$$

$\frac{2\pi}{5} < x < \frac{4\pi}{5}$



2) $\sin x < 0$ $\quad \pi < x < 2\pi$

$$\frac{4\sin^2 x - 3}{\sin x} > \frac{2\cos x}{\sin x} \quad (\cdot \sin x)$$

$$\frac{4\sin^2 x - 3 + 2\cos x}{\sin x} > 0$$

$N > 0 \quad 4(1-\cos^2 x) - 3 + 2\cos x > 0$

$D > 0 \quad \sin x > 0$

$N > 0 \quad -4\cos^2 x + 2\cos x + 1 > 0$

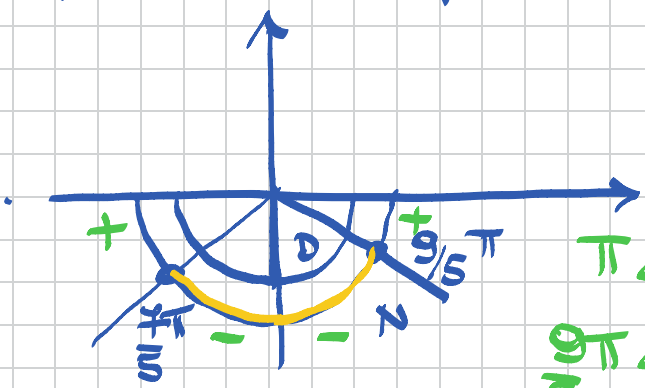
$4\cos^2 x - 2\cos x - 1 < 0$

$\cos x = \frac{1 \pm \sqrt{5}}{4}$

$\cos x = \frac{1-\sqrt{5}}{4} \quad x = \frac{2\pi}{5}; \; x = \frac{4\pi}{5}$

$\cos x = \frac{1+\sqrt{5}}{4} \quad x = \frac{\pi}{5}; \; \frac{3\pi}{5}$

$$\frac{1-\sqrt{5}}{4} < \cos x < \frac{1+\sqrt{5}}{4}$$



$\pi < x < \frac{11}{5}\pi$

$\frac{9}{5}\pi < x < 2\pi$